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Assignment No: 4

Problem Statement:

Build a simple question-answering chatbot using a pre-trained language model.

Objective:

- To understand the extractive question-answering task and the mechanism by which transformer models solve it.
- To load Hugging Face's QA pipeline backed by a pre-trained BERT-based model.
- To supply a structured knowledge context as the chatbot's information source.
- To retrieve precise answer spans from context passages using natural language questions.
- To wrap the system in an interactive input loop that accepts user queries until an exit command is entered.

S/W Packages and H/W apparatus used:

- Operating System: Windows/Linux/macOS
- Kernel: Python 3.x
- Tools: Google Colab / Jupyter Notebook
- Hardware: CPU with minimum 4GB RAM; GPU recommended for faster inference
- Libraries and packages used: Transformers (Hugging Face pipeline, DistilBERT-base-cased-distilled-SQuAD)

Theory:

Question Answering (QA) is an NLP task in which a system locates and returns the most relevant answer to a natural language question from a given information source. QA systems fall broadly into open-domain QA — retrieving answers from large corpora — and extractive or closed-domain QA — identifying answer spans from a fixed context passage. This assignment targets extractive QA, where answers must be verbatim substrings of the supplied context.

Extractive Question Answering:

The model receives a question Q and context C as a joint input and predicts start and end token positions that delimit the answer span. Two probability distributions P_{start} and P_{end} are computed over all N context tokens, and the answer is extracted as the text from $\text{argmax}(P_{\text{start}})$ to $\text{argmax}(P_{\text{end}})$. This span-extraction approach ensures answers remain grounded in the context, eliminating any possibility of hallucinated facts.

DistilBERT for Question Answering:

The pipeline defaults to distilbert-base-cased-distilled-SQuAD — a knowledge-distilled version of BERT fine-tuned on the SQuAD reading comprehension benchmark. DistilBERT achieves 97% of BERT's accuracy while being 40% smaller and 60% faster. SQuAD's 100,000+ Wikipedia-sourced QA pairs make this model highly reliable for factual question answering across general domains.

BERT's Bidirectional Encoder:

BERT's Transformer encoder attends to all tokens bidirectionally in a single pass, providing deep contextual understanding. For QA, the question and context are concatenated with a [SEP] separator into one input sequence. Two linear heads placed on the final hidden layer predict the start and end answer positions respectively — a selection mechanism fundamentally distinct from autoregressive generation.

Hugging Face QA Pipeline:

The pipeline('question-answering') callable bundles tokenization, model forward pass, span extraction, and decoding into a single API call. It returns a dictionary containing the extracted answer string, a confidence score, and character-level start and end positions within the context, making deployment straightforward without manual model management.

Methodology:

- Install the Hugging Face Transformers library.
- Import the pipeline function from transformers.
- Initialise the QA pipeline using pipeline('question-answering'), which auto-downloads DistilBERT-SQuAD.
- Define a four-sentence context covering Artificial Intelligence, Machine Learning, Deep Learning, and NLP.
- Submit the test question 'What is Machine Learning?' to the pipeline with the context and print the answer.
- Implement an interactive while-loop accepting user questions via input(), querying the pipeline each iteration, and breaking on 'exit'.

Use Cases:

- Use Case 1 - Internal Knowledge Base Bot: Companies deploy QA chatbots over HR manuals, policy documents, and technical wikis, allowing staff to retrieve specific answers without manually searching lengthy documents.
- Use Case 2 - Student Study Assistant: Learners load textbook passages as context and ask conceptual questions, receiving context-grounded answers that support exam preparation and self-directed study.
- Use Case 3 - Automated Customer Support: Product FAQ documents serve as the context layer, enabling e-commerce platforms to resolve common queries automatically without human agent intervention.

Advantages:

- Zero training required — DistilBERT-SQuAD delivers high-quality QA immediately upon loading.
- Extractive grounding guarantees that all answers are verbatim context substrings, preventing hallucination.
- The pipeline API reduces the full QA system to effectively two lines of code.
- The interactive loop is readily extensible into GUI or web-deployed chatbot interfaces.
- DistilBERT's 66M parameters enable practical CPU inference without GPU requirements.

Limitations:

- Only questions whose answers are explicitly present in the context can be answered.
- Contexts exceeding 512 tokens trigger truncation that may discard relevant information.
- No conversational memory — each query is processed independently with no cross-turn context.
- Multi-hop reasoning and synthesis across non-adjacent context sentences degrade answer quality.
- Static context requires manual updates when the underlying knowledge base changes.

Applications:

- Legal research: Extracting specific answers from case law and statutory documents.
- Clinical decision support: Answering physician queries from treatment guidelines and drug information.
- Government e-services: Citizens query eligibility criteria from policy documents in plain language.
- Developer tooling: Conversational querying of API documentation and troubleshooting guides.
- Journalism research: Rapidly extracting key facts and quotations from lengthy news articles.

Conclusion:

This assignment built a context-aware extractive QA chatbot using Hugging Face's DistilBERT-SQuAD pipeline. A four-sentence AI knowledge context was defined as the information source. The system accurately identified 'a subset of AI that allows computers to learn from data' as the answer to 'What is Machine Learning?'. An interactive query loop was implemented to support continuous user sessions. The experiment demonstrates how pre-trained transformer QA models can be deployed as functional, hallucination-free chatbots with minimal engineering, making extractive QA broadly accessible across domain-specific knowledge retrieval applications.